Prospective evaluation and component attribution of hypothesis-space expansion in AI systems

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Abstract

A hypothesis representation can limit what an AI system finds, even when search within that representation is effective. Modern discovery systems complicate evaluation because language-model proposals interact with programmed transformations, realizers, fitters and selection policies. We introduce a prospective executable assay for bounded hypothesis-space expansion and couple it to component-level provenance. A candidate must leave a frozen structural language, fit observations, commit to a discriminating intervention before outcome reveal, improve on a selected observation-optimal incumbent and survive held-out falsification. In 400 synthetic worlds, external typed proposals produced validated escapes in 142, compared with one each for direct and additional-sampling model conditions. Structured composition succeeded in all 400 known-family worlds and 100 worlds from one search-side held-out family, under a hand-authored, family-aligned realization library that already contained the held-out basis. A post-hoc inference-free replay reproduced all 2,400 candidate verdicts: deterministic components supplied the evaluated theories, ranking and interventions. Conversely, historical self-composition failed before executable evaluation. A grammar-constrained interface exposed model-authored graph proposals, yielding successes in 15 of 96 worlds versus 16 for random composition on the same panel. The resulting evidence records distinguish validated system escape, model-authored content and dependence on supplied explanatory forms. This operationalizes a longstanding representation-search distinction while showing how prospective validation and artifact-level attribution can jointly identify what a hybrid system achieved and which components supplied it.

Keywords: knowledge representation; theory revision; heuristic search; abduction; hypothesis-space expansion; component attribution

1. Introduction

Searching a hypothesis language and changing that language are different operations. A learner can search effectively yet fail to reach a useful theory because its representation makes that theory unavailable, cumbersome or inaccessible to its revision operators. Constructive induction, predicate invention and theory restructuring have treated this problem for decades [7-9,32,40]. The distinction remains relevant when a language model proposes hypotheses: what appears to be a new explanation may originate in a model-generated representation, a program transformation, or a realization rule supplied by the system designer.

Recent discovery systems combine several such processes. Symbolic regression searches executable expressions; program-search systems combine learned proposals with deterministic evaluation; scientific agents retrieve literature, propose mechanisms, execute tests and revise hypotheses [10-20]. Some explicitly expand principle spaces, mechanistic model classes or causal structures [23-26]. Meanwhile, agent-evaluation research shows that parsers, tool interfaces and execution scaffolds can change scores in both directions [36-38]. A system-level outcome therefore leaves two questions unresolved: did the evaluated candidate change the declared representation, and who supplied the content responsible for its predictions?

These questions require more than a novelty judgment or an ablation of the model name. A new graph may be paired with an equation selected from a fixed basis library. A model may produce text that is replaced before evaluation. An apparently unsuccessful proposal may never reach an executable interface. Conversely, a model-authored expression can enter the evaluated artifact even when its representation was supplied externally. Each path supports a different claim about the system.

We study these distinctions through **bounded hypothesis-space expansion**. The term denotes a conjunction of structural and predictive requirements, not a measure of abduction or creativity as a whole. The incumbent structural language and finite executable comparator set are frozen. A candidate must be structurally outside that language, fit public observations and make a committed prediction that distinguishes it from the selected observational comparator. Hidden intervention outcomes and separate falsification cases then determine whether the candidate's predictive advantage survives. Structural non-membership is recorded independently of predictive performance.

The contribution has three parts. First, we specify a prospective executable evaluation target, with explicit separation between a representation language, a finite comparator set and the observation-optimal comparator selected from it. Second, we use frozen experiments to distinguish within-language controls, atomic representation proposals, local rewrites and complete compositional search policies. Third, we connect each verdict to a field-level provenance and replay procedure that records supplied knowledge, accepted model fields and overwrites. The procedure identifies both successful artifacts whose decisive content is deterministic and apparent failures caused by pre-execution attrition.

The experiments provide a deliberately exact setting for this analysis. Structured search saturates the studied generators under an aligned, hand-authored realizer. Removing model output preserves its verdicts. Repairing a different model-proposal interface yields executable graph edits but no aggregate advantage over random composition. These are informative results about the implemented paths, not a competition over general reasoning ability. Their methodological value is that the same end-to-end scoring rule can be joined to evidence explaining what was actually evaluated.

2. Background and related work

2.1. Abduction and theory revision

Peircean hypothesis formation and inference to the best explanation concern explanatory reasoning, but neither supplies a universal computational test of creativity [1-3]. A predictive theory may be useful without being structurally new, and a structurally novel proposal may explain nothing. Our assay operationalizes one bounded component of abductive reasoning: proposing an executable representational alternative and testing a discriminating consequence prospectively. The separation of observational agreement from interventional prediction follows the causal-inference distinction between those two sources of evidence [30,31].

Theory revision also includes several distinct operations. Belief update and revision can change accepted sentences within a fixed logical vocabulary; their semantic characterization is different from changing the vocabulary itself [42]. The ABC repair approach explicitly distinguishes adding axioms, deleting conflicting axioms and reforming a theory's language [41]. Timber assembles runnable qualitative models from fragments, organizes competing explanations and revises beliefs through metareasoning [33]. These precedents make representation and explanation construction central technical objects. Our contribution is an evaluation contract linking structural escape, prospective prediction and component provenance, rather than a new general account of theory change.

2.2. Constructive induction and representation change

Predicate invention introduces useful predicates beyond those initially supplied [7,8]. Constructive induction generates descriptors that change how examples and hypotheses are represented [40]. Donoho and Rendell distinguish limitations of a representation language from limitations of a theory's structure, showing why local revisions can be inadequate [9]. Meta-interpretive learning further provides executable predicate invention and recursive program induction under metarules [32]. These methods provide the foundations for representation-changing search examined here.

Boden's exploratory/transformational distinction and Wiggins's formal accounts of creative spaces provide a related conceptual vocabulary [4-6]. We use a narrower operational boundary. Whether a graph is admitted by a declared structural language is decidable in our implementation; whether its predictive function is inexpressible by every possible incumbent expression is a different question. Our membership certificate addresses the former.

Open-world AI gives this distinction a system-level setting. HYDRA repairs environment models through heuristic search [34], while a neurosymbolic architecture evaluates both components and integrated novelty accommodation [35]. Recent online causal-model learning uses dynamic predicate invention in a predict-verify-refine loop [39]. Their evaluation targets adaptation to changing environments; ours is the provenance of the particular executable artifact receiving a prospective escape verdict. Representation repair and component evaluation already coexist in these research programs.

2.3. Search-based AI discovery

Symbolic regression and physics-informed equation discovery search expressive program spaces with quantitative evaluators [10,11]. Algorithm discovery can similarly use reinforcement learning or language-model-generated programs [12,13]. In FunSearch, the language model proposes code and an evaluator guides selection [13]. This division of labor is explicit; it should not be collapsed into an undifferentiated claim of model reasoning.

The choice of language remains relative to the system being studied. A search procedure can leave a restricted incumbent language while remaining inside a larger designer-supplied meta-language. Our compositional experiments have exactly this character: graph rewrites trigger a fixed motif-to-basis realizer. They test how search organization accesses supplied predictive forms. Their success is compatible with a wholly deterministic path and with the decisive algebra having been specified in advance.

2.4. Language-model hypothesis generation and scientific agents

Hypothesis Search compiles hypotheses into executable programs, and other language-model approaches iteratively generate and update hypotheses for predictive tasks [14,15]. POPPER designs and executes sequential falsification tests with statistical error control [16]. Broader systems automate research workflows, integrate scientific tools, or validate selected proposals experimentally [17-20]. These contributions concern different combinations of proposal quality, execution, validation and workflow breadth.

Recent work is not uniformly restricted to a static hypothesis set. PiEvo evolves a principle space [23]. Model Discovery Agent couples model-proposed mechanistic structures to Bayesian fitting and experimental design, reports an expansion-component ablation and explicitly separates the proposer from numerical inference [24]. EvoSCM revises causal structures and mechanisms through discriminating interventions and committed predictions [26]. HypoArena evaluates open-ended hypotheses from reconstructed pre-conclusion contexts using reference separation and structured judging [25]. Its temporal construction is different from committing an executable candidate before revealing a simulator outcome.

HypoSpace and ResearchBench provide complementary diagnostics for underdetermination and task decomposition [27,28]. Creativity-task scores and historical discovery reconstructions likewise measure useful but distinct outcomes [21,29]. Zahavy's position motivates the question of generating explanatory premises [22]; our assay evaluates a bounded, executable instance of representational change.

2.5. Evaluation and attribution of hybrid systems

Harness-aware evaluation directly challenges treating agent scores as properties of model weights alone [36,37]. Standardized interfaces and environments expose failures caused by parsers or execution conventions, while complete-trace benchmarks study the responsible agent and decisive step of a failure [38]. We adopt this concern with observability but target a particular object: the explanatory content of a prospectively evaluated theory.

Harness-level evaluation asks how system configurations affect scores, whereas failure attribution identifies where an execution went wrong. Our target is the committed explanatory artifact: which components supplied its representation, predictive expression and intervention, and what a specified replay intervention changes in its prospective verdict. This distinguishes evaluation targets, without claiming that adjacent work excludes artifact provenance. The concrete record in Supplementary S20 links raw responses to the scored theory and its replay.

3. Formal framework

3.1. Incumbent language and executable comparator

A world supplies public observations D_{obs}, a finite set of public intervention actions U with their input settings, an incumbent representation R_0 and a frozen structural language A_0. Hidden information consists of intervention outcomes and a separate falsification split D_{fal}. Public actions expose possible inputs, not their outcomes. The procedural generator and evaluator can access hidden truth; proposers, fitters and ranking procedures cannot.

Let P_0 be the frozen finite set of incumbent executable programs. A_0 specifies admissible structures; P_0 specifies the actual executable comparators examined. They are not interchangeable. For a program h, $L_{obs}(h)$ is mean squared error on D_{obs}. The selected observation-optimal comparator is

$$h_0^* = \arg\min_{h \in \mathcal{P}_0} (L_{\mathrm{obs}}(h), \mathrm{canonical}(h))$$

The ordered pair is minimized lexicographically, with canonical program JSON breaking observation-loss ties. Intervention outcomes do not select the comparator. Exhaustiveness applies to P_0, not to every function expressible under A_0. Nor do we assume that all observation-optimal programs make identical interventional predictions. All reported gains refer to the selected h_0^*.

3.2. Candidate theories and structural escape

A candidate theory is T=(R,f), where R is a typed structural representation and f is its executable predictive expression. Representations record nodes, edges, equations, dependencies, constraints, observability, temporal indices, arities and argument bindings. Canonical serialization sorts unordered fields. Membership checks inspect the frozen structural constraints and retain concrete reasons for rejection.

Structural escape requires a valid representation $R \notin A_0$. Renaming or paraphrasing cannot by itself satisfy this test. The certificate concerns structural non-membership; it is not a proof of functional inexpressibility over the whole incumbent language. Because R and f are separate objects, provenance must record how a representation becomes a predictive expression. A motif realizer can supply f even when another component supplied R.

3.3. Prospective commitment and six gates

Before outcome reveal, the system commits a record containing the canonical candidate, expression, selected action, predicted outcome and data-split hashes. For the compositional paths, the action selector uses candidate and comparator predictions on public inputs:

$$a^* \in \arg\max_{a \in \mathcal{U}} |f(a) - h_0^*(a)|$$

The deterministic implementation resolves ties reproducibly. This is outcome-blind prediction separation, not selection using the realized gain. Let L_{int} denote error on the committed intervention evaluation and L_{fal} error on the separate falsification cases. Define $G_{int}=L_{int}(h_0^*)-L_{int}(f)$ and $G_{fal}=L_{fal}(h_0^*)-L_{fal}(f)$. Let V(T) denote joint validity of the graph, executable expression and bounded feature-support checks. Supplementary S19 lists the exact checks and their limits; they do not establish graph/expression semantic equivalence. The implementation conjoins V(T) with each of J1-J5; the table shows the additional gate-specific requirements. The frozen gates are:

Gate	Executable requirement	Role
J0	$L_{obs}(h_0^*) \leq 10^{-12}$	Incumbent already explains the public observations
J1	$R \notin A_0$, with $V(T)$	Structural escape with a membership-failure record
J2	$L_{obs}(f) \leq 10^{-12}$	Candidate also explains the observations
J3	Committed prediction separation ≥ 0.5	Candidate and selected comparator discriminate before reveal
J4	$G_{int} > 0.1$	Candidate improves on the selected comparator after reveal
J5	$L_{fal}(f) \leq 10^{-12}$ and $G_{fal} > 0.1$	Exact survival and gain on separate falsification cases

Table 1. Six-gate evaluation contract. J3 uses committed predictions before outcome reveal; J4 and J5 use hidden outcomes.

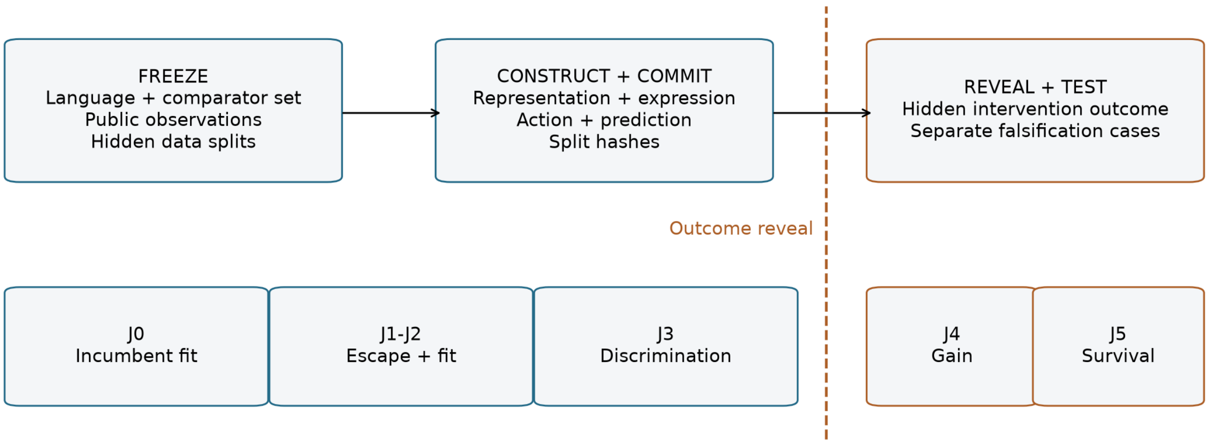

Figure 1. Prospective evaluation boundary. Language, comparator rule and data splits are frozen before construction. The candidate and intervention prediction are committed before outcome reveal. Schematic prepared with ChatGPT/Codex assistance and rendered programmatically; no generative image model was used.

The bounded expansion event is $J(T)=J0 \wedge J1 \wedge J2 \wedge J3 \wedge J4 \wedge J5$. A world succeeds if at least one of its three retained candidates passes J. The jump success rate is the proportion of jump worlds that succeed. Candidate counts describe attrition; worlds are the inferential units.

This conjunction makes several distinctions executable. Retrospective fit alone satisfies neither structural escape nor prospective discrimination. Structural escape alone supplies no predictive advantage. A fixed-language control cannot pass J1, regardless of its search quality. If the selected comparator is exact on every scored hidden control case, positive error reduction is impossible; a zero control rate then checks construction-specific consistency rather than estimating a general false-positive rate.

The gates are an operational contract for these noiseless worlds. The thresholds are not a statistical error-control theorem. No semantic creativity score, explanation quality, model confidence or embedding distance enters J.

3.4. Provenance and outcome dependence

For every evaluated field, an evidence record stores the producer, parsed model value where applicable, transformations, final value and whether the evaluator reads it. At minimum, the record covers representation, expression, fitted parameters, ranking and intervention. Overwrites are retained as transformations rather than silently assigning the final artifact to the original proposer.

The replay intervention specifies a replacement and what remains fixed. Given an archived artifact c, an evaluator E and a field intervention I, equality E(c)=E(I(c)) supports invariance under that intervention in the tested path. It does not imply that the component is irrelevant under every alternative design. Conversely, a changed verdict localizes dependence on the replaced content, but its interpretation depends on the replacement: substituting the comparator expression necessarily removes J3 separation and is a structural negative control.

The reusable audit combines five records: (1) the pre-reveal commitment, (2) field-level provenance, (3) executable replay results, (4) the intervention specification and (5) an inventory of supplied knowledge. The inventory includes operator semantics, basis libraries, fitting procedures, search/ranking rules and action-selection policy. Together they separate the claims in Table 2.

Claim	Required evidence beyond a headline score
System-level validated escape	Frozen language and comparator rule; committed executable theory/action/prediction; all six gates
Model contributed the evaluated representation	Model-authored structural content survives parsing and transformations into the scored representation
Model content affects success	A specified field/component intervention changes the evaluated outcome, with fixed components and replacement semantics reported
Origin of the explanatory form	Identify whether the form was directly supplied as a template, constructed compositionally or generated by a learned proposer; document the supplied library and derivation
Autonomous theory revision	The claimed autonomous boundary covers proposal, realization and validation; external theory content and human interventions are explicitly accounted for

Table 2. Evidence required for different AI discovery claims. Records of origin and construction do not by themselves establish scientific novelty. Construction within supplied primitives is compatible with invention; exceeding the expressive capacity of the entire meta-language is not a necessary criterion. Ordinary authorship of a field and causal dependence of the success indicator are different estimands.

Procedure box. Freeze the evaluated object and public/hidden contract; trace each scored field through parsing, realization, fitting and overwrites; label runtime records separately from code-inspected or reconstructed provenance; specify the replacement and fixed components; recompute the commitment and gates; report the changed fields, verdict and supplied forms. Schema schemas/aij-evidence-record-v1.schema.json and tools scripts/build_aij_evidence_record.py and scripts/validate_aij_evidence_record.py implement record version 1.0.0. Supplementary S20 illustrates the procedure with the archived candidate used in Section 5.1.

4. Experimental framework

4.1. Procedural worlds and freezes

AJ5 contains eight procedural families: latent common cause, unification, hidden regimes, property-to-relation, state invention, coordinate transformation, causal ambiguity and meta-law. Each has 50 jump worlds and 25 control worlds. CJ5 uses new seeds for the same family counts and adds 100 jump and 100 control worlds from triadic relation reification. The shared benchmark design is synthetic and noiseless. Its purpose is exact verification of public/hidden boundaries, structural membership and predictions.

World generation uses no model output. The generator supplies observations compatible with an incumbent program and cases on which an escaped candidate can discriminate. Controls retain an incumbent-compatible truth. Generator validation checks deterministic generation, hidden-truth redaction, observational adequacy and bounded constructive reachability.

The historical protocols were specified and frozen in Git before confirmatory model calls. These public commits are not independent registry timestamps; we describe the experiments as commit-frozen, not formally preregistered. Later interface and realizer analyses have their own freezes and are post-confirmatory sensitivities or audits. No main experiment was rerun for this manuscript. Supplementary Sections S1-S11 identify seeds, freezes and replay artifacts; S12-S17 distinguish completed sensitivities, operational amendments and excluded partial runs.

4.2. AJ5 search-space decomposition

B0 directly emits a typed representation. B1 increases independent sampling at higher temperature. B2 retains the incumbent representation while permitting within-space reasoning. B3 changes attributes or values within the incumbent language. B4 samples three proposals with replacement from a nine-member typed transformation portfolio. B5 selects three structurally distinct portfolio proposals with archive accounting and deterministic falsification. A separate value-only control also remains within the incumbent language.

B2/B3 and the value-only condition are structural controls, not fair contests over universal capability: their reachable representations cannot satisfy J1. B4/B5 are high-level proposals aligned to the procedural mechanisms. AJ5 asks whether representation-level inputs reach validated escape under this supplied proposal space.

A source-factorial comparison uses model-proposed, external-portfolio and supplied-correct representations in the same downstream path. The fitter and maximum-separation action selector supply values shown to the second model call and enforce them in the evaluated artifact. This comparison estimates the effect of the representation source on the complete pipeline, not the necessity of downstream model reasoning.

4.3. Generic rewrites, realization and search

CJ5 replaces atomic transformations with 29 local graph/syntax rewrites. Typing a newly added node, changing observability or arity, and binding arguments require separate edits. Operations do not receive family names, target distance or hidden outcomes. An ancestry record binds parent and child hashes, operation, canonical arguments, seed and depth.

C0 evaluates within-space candidates. C1 retains the high-level atomic reference. C2 evaluates depth-one generic alternatives. C_rand samples 48 four-step paths and retains candidates by seed-fixed structural hash. C3 traverses 48 four-step branches, with fixed allocations among generic topological patterns. It evaluates each intermediate graph through the realizer and fitter, then scores validity, escape, observational compatibility, public-action separation and diversity. Its final diversity-aware ranking retains three terminal candidates. Thus fitting precedes scoring and final selection, not only the final commitment.

C3 and C_rand share primitives and a realizer but differ in both traversal and final selection. Their contrast is between complete search-and-selection policies. C_self requests 16 four-step plans for each of three slots from the model; invalid plans consume their capacity. Conditional ceilings and unequal-semantics references are reported separately in the supplement.

The realizer matches graphs to nine hand-authored signatures and fits coefficients in their fixed basis libraries. It does not learn an unrestricted graph-to-law mapping. The supplied inventory is:

Signature	Supplied basis/content	Aligned mechanism
Arity-three relation	$x \cdot z \cdot w$	Triadic held-out product
Temporally indexed recurrence	x and history sums	State invention
Unobserved dependency	Stable observed proxy	Latent common cause / causal ambiguity
Unobserved selector	Regime-signed x	Hidden regimes
Bound relation	Observed scalar variables	Property-to-relation
Self-composed function	x^2	Coordinate transformation
Shared-rule binding	x and context	Unification
Multi-argument function	x and $x \cdot$context	Meta-law
Incumbent fallback	First two scalar variables	Fallback incumbent form

Table 3. Supplied realization inventory. Graph signatures select hand-authored basis content before observation-only coefficient fitting.

The triadic basis was present before held-out unlock. The held-out family therefore tests search-side structural transfer to one adjacent arity-three construction, not discovery of an unseen algebraic form. The language already contains reification and argument binding, and AJ5 includes a binary property-to-relation family.

4.4. Models and interface sensitivities

Historical experiments use frozen microsoft/phi-4, revision 2db69c1c3e91a05d2c64a3185acfbaf36f744e25, served by vLLM 0.10.2 with dynamic bitsandbytes 4-bit quantization, a 4,096-token context and a 700-token completion cap. Temperature is 0.2 except AJ5 B1 sampling and factorial P0 at 0.7; top-p is 0.95. Requests use deterministic seeds; no fine-tuning or cross-world adaptation occurs. Allocations are reported separately from realized token use.

The targeted sensitivity panel selects 12 existing seeds per known family by outcome-blind SHA-256 ranking, giving 96 paired worlds. Legacy-interface conditions vary precision/serving, output budget, model or one validator-only repair. A separate 40-world positive control supplies the correct representation while evaluating the model-authored expression and action without overwrite. It differs from the historical source factorial.

The DeepSeek service identifies deepseek-ai/DeepSeek-V4-Flash-Vision-Exp, revision 86f746b36186f0e567729a5c06a8c918caba82a9, on the patched vLLM runtime recorded in Supplementary S15. We do not substitute a different public model name. Native reasoning is returned separately from the final parser answer. The native and increased-budget conditions are not compute-matched to the historical cap.

The grammar-constrained sensitivity uses the same 96-world panel and 48 four-step opportunities per world. Each slot receives a native-reasoning call followed by a serialization call with exact operation keys and a constrained output schema, each capped at 4,096 tokens. This replaces the proposal-plus-explanation allocation, preserving six calls per world, but changes interface and token allocation together. Without a donor graph, 28 non-crossover primitives are available. Schema validity guarantees format, not dynamic executability or predictive adequacy.

Neither call receives truth, target distance, fitted expression, hidden outcome or gate feedback. The model supplies graph-edit topology. The unchanged deterministic scaffold supplies realization, fitting, ranking and the committed intervention. Full prompts, errors and operational exclusions are retained in the supplement and archive.

4.5. Component interventions and statistical analysis

The post-hoc C3 audit removes both model outputs, substitutes a valid empty explanation, reconstructs the deterministic artifact and recomputes commitments and gates without inference. It tests the implemented content path, not a retrained or redesigned system.

The separately frozen realizer audit fixes 3,288 archived candidate slots from C3/C_rand on 500 jump worlds and the grammar-constrained condition on 96 worlds. Eleven policies include aligned replay, incumbent-expression substitution, role/action-blind rebinding and eight individual non-incumbent signature masks. Blind rebinding preserves motif algebra but assigns type-compatible fields lexically, without node-role or intervention-change information, then refits and reselects the action outcome-blind. A signature mask falls back to the incumbent expression only for that signature. Graphs and slots stay fixed; adaptive re-search is outside the intervention.

World-level counts and rates are primary. AJ5 uses family-stratified paired bootstrap intervals from 10,000 replicates. Its archived uncentered bootstrap tail quantities are not null-calibrated P values and are not used for significance claims. CJ5 also reports paired sign-flip tests with correction within frozen comparison families in the supplement. Candidate rows describe attrition, never independent replication. Paired sensitivities use identical worlds and disclose joint outcomes. Family rates are descriptive: many seeds within co-designed generators do not establish breadth over independently authored domains.

5. Results

5.1. The assay separates fit, structural escape and prospective discrimination

The worked triadic world makes the distinction concrete. Public observations satisfy $x=z=w$ and $y=9x^3$. Both the selected incumbent $y=9x^3$ and the candidate $y=9xzw$ fit those observations exactly. Four generic edits reify the prediction edge, increase relation arity and bind z and w. The supplied realizer contributes the xzw basis and fits its coefficient.

Before outcome reveal, the system commits to changing z from 6 to 7 while $x=w=6$. The incumbent predicts 1,944 and the candidate 2,268; the revealed value is 2,268. A separate falsification case at $z=5$ yields 1,620, again matching the candidate. Structural membership fails on the recorded relation constraints, and all six gates pass. The example demonstrates validated escape under a supplied algebra, not algebraic invention.

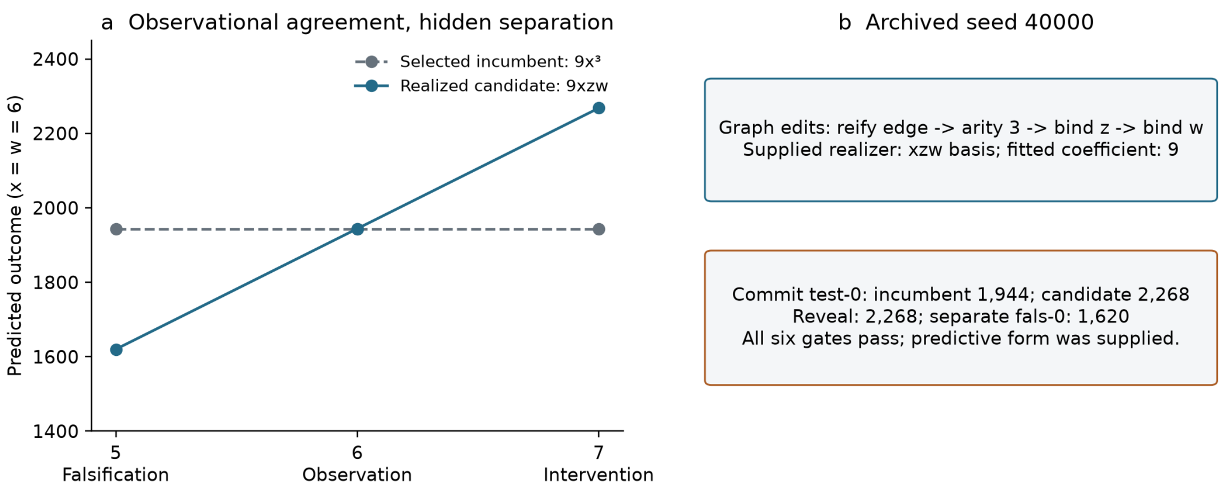

Figure 2. Reconstructing the worked prospective example in Section 5.1. Archived held-out seed 40000 supplies the observations, committed test-0 and separate fals-0 outcome. The plotted slice fixes $x=w=6$. The complete falsification split also includes a second case, used in the all-gate verdict. The graph leaves the structural language; its product basis is supplied by the realizer. Figure prepared with AI-assisted code from the archived candidate and deterministic world reconstruction.

Controls show why the prospective gates matter. Among 900 C3 control candidates, 800 pass J1, 567 pass J2 and 283 pass J3, yet none passes J4. In AJ5, B4/B5 likewise retain 112/110 control candidates through J3, with none surviving J4. Thus fit plus structural departure plus predicted separation is insufficient.

An offline check verifies that the selected comparator exactly matches every scored control outcome: 1,125 intervention/falsification cases in 200 AJ5 controls and 1,625 in 300 CJ5 controls. Under this construction no candidate can achieve strictly positive error reduction. The zero control verdicts are therefore construction-specific checks of the assay, not a measured zero false-positive rate for noisy scientific discovery.

5.2. Representation-level proposals cross the declared boundary

AJ5 condition	Reachable representation input	Selection / control role
B0 direct model	Model-emitted typed graphs	Direct proposal path
B1 additional sampling	Model-emitted typed graphs	Higher-temperature sampling
B2 fixed-space reasoning	Incumbent graph only	Structural control: J1 unreachable
B3 attribute mutation	Within-language attributes	Structural control: J1 unreachable
B4 typed portfolio	Nine supplied transformations	Three draws with replacement
B5 distinct portfolio	Same supplied transformations	Three structurally distinct proposals; archive accounting

Table 4. AJ5 reachable inputs and selection roles; Figure 3 reports counts and uncertainty.

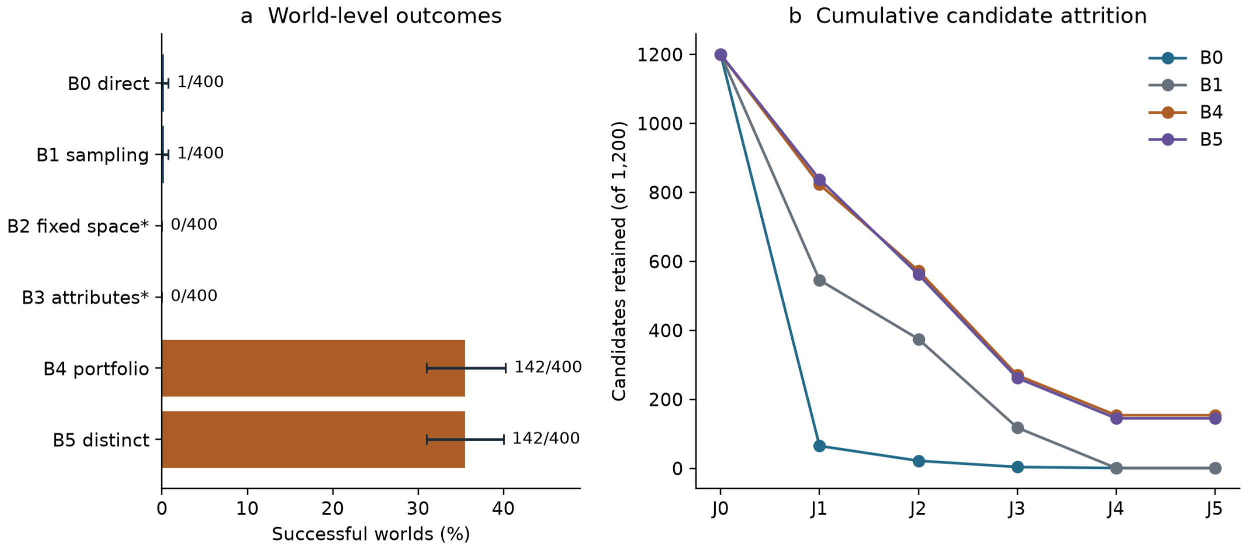

Figure 3. Representation-level proposals under the prospective assay. (a) World success counts, n=400 per condition. Asterisks mark fixed-language structural controls whose reachable graphs cannot pass J1. (b) Cumulative gate retention among 1,200 candidates per condition; candidates are not independent replicates. Whiskers show the existing 95% family-stratified bootstrap intervals. Figures 3, 4, 6 and 7 are reproducible plots of frozen artifacts, prepared with AI-assisted plotting code.

Both external proposal conditions achieve 35.5% world success. Their shared aggregate does not imply identical candidate trajectories. Among 1,200 jump-world candidates each, B4 retains 823 at J1, 573 at J2, 270 at J3 and 154 at J4-J5; B5 retains 838, 562, 262 and 145. Direct proposals retain 65, 22, 4 and 1; additional sampling retains 546, 375, 118 and 1. These executed proposal paths fail at different stages, which a world-level rate alone conceals.

The representation-source factorial yields 0/400 for model-proposed representations, 142/400 for the external portfolio and 400/400 for supplied-correct representations. Its downstream fitted expression and intervention are enforced by the scaffold. The result shows sensitivity to representation input in that path; it does not identify a necessary contribution from the second model call.

5.3. Search organization matters under the fixed realizer

CJ5 policy	Reachable construction	Selection / control role
C0 fixed space	Incumbent language	J1 structural control
C1 atomic reference	High-level supplied transformations	Atomic-proposal reference
C2 depth one	One generic edit	Local-rewrite control
C_rand	48 four-step sampled paths	Seed-fixed structural-hash selection
C3 structured	48 four-step structured branches	Intermediate fit/separation scoring; diversity-aware retention
Historical C_self	Requested model four-step plans	Best evaluated plan per slot; historical pre-execution attrition

Table 5. Reachable constructions and selection policies under the fixed realizer; Figure 4 gives exact counts.

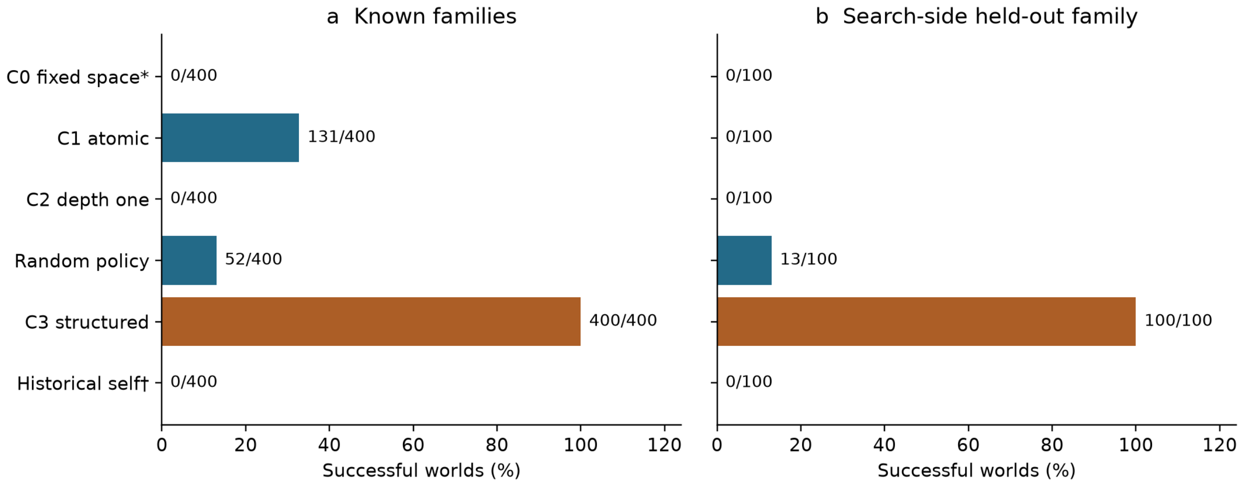

Figure 4. Generic search under a fixed realization language. Known-family and search-side held-out world counts have different denominators. C0 (*) is a structural control; historical self-composition (†) is interface-confounded. The C3/random contrast includes traversal and selection, not proposal source alone. The held-out algebraic basis was already supplied.

C3 saturates the known-family generators and the single search-side holdout. Its paired success-rate difference from C_rand is 0.87 in each population, with bootstrap 95% intervals [0.845, 0.895] and [0.80, 0.93], respectively. This estimates a complete-policy contrast: organized traversal and outcome-blind fitted ranking together access successful motif triggers more reliably than random paths with structural-hash selection.

The held-out success is qualified by the supplied inventory, not by an after-the-fact exception. The search-side family was locked during known-family confirmation, but the realizer already supplied the triadic product. The result establishes transfer of a bounded graph-search procedure to an adjacent structure under an existing realization language.

5.4. Field provenance and replay assign C3 success to the scaffold

In C3, representation construction, realization, fitting, ranking and action selection occur deterministically before the model response. Model-proposed representation, expression and intervention fields are overwritten. The remaining explanation is not read by J0-J5.

Evaluated object	C3 source	Grammar-constrained source	Supplied-representation sensitivity source
Representation	Deterministic graph search	Executed model graph edits	Externally supplied correct graph
Executable expression	Motif realizer	Motif realizer	Model-authored expression
Parameters / fit	Deterministic observation fit	Deterministic observation fit	Values in the model-authored expression
Candidate ranking	Deterministic policy	Deterministic policy	Distinct supplied-representation control path
Committed action	Deterministic maximum separation	Deterministic maximum separation	Model-selected action
Explanation	Model text, unscored	Not a scored theory field	Model text, unscored

Table 6. Final evaluated-field provenance differs across implemented paths.

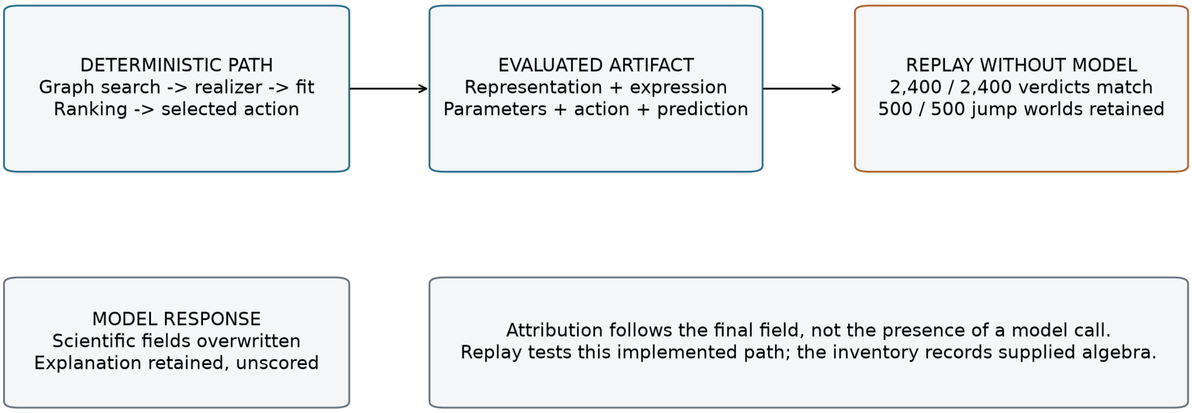

Figure 5. C3 evaluated-field provenance and post-hoc replay. Scientific model fields are overwritten before scoring; the retained explanation does not enter the gates. Removing model outputs preserves all candidate verdicts and all jump-world successes, alongside zero control successes. Schematic prepared with ChatGPT/Codex assistance and rendered programmatically.

Removing the C3 model outputs reproduces all 2,400 candidate verdicts, retains all 500 jump-world successes and retains 0/300 control successes. This post-hoc intervention establishes that evaluated C3 success is independent of the removed model text in this implemented path. The successful artifact is authored by typed search, the supplied realizer, fitting and selection.

Provenance and replay answer complementary questions here. The provenance trace identifies the overwritten fields; replay checks that reconstructed commitments and evaluation preserve the archived outcomes.

5.5. Interface attrition separates non-execution from tested proposals

Historical C_self failures occur before executable plan evaluation. All 1,200 known-family proposal responses are non-empty and the legacy parser extracts a JSON object, but none has an outer plans list. All reach the 700-token cap, and none of 19,200 plan opportunities executes. Object extraction is therefore a misleading success indicator for this interface.

The legacy prompt omits exact executor argument keys. Targeted legacy-interface sensitivities remain at 0/96, including matched and native DeepSeek paths; native reasoning appears in a separate field without a usable final parser answer. These observations identify an execution boundary, not the absence of an internally useful conceptual proposal.

The two-stage grammar-constrained interface makes all 4,608 opportunities schema-valid and 3,939 dynamically executable. At least one executable plan survives in 280/288 slots; 236 selected candidates pass J1-J2 and 21 pass J3-J5, yielding 15/96 successful worlds (15.6%; existing Wilson 95% interval 9.7-24.2%).

Successes occur in meta-law (9/12) and unification (6/12). The model thus supplies evaluated graph topology, while the realizer still supplies the predictive law.

Random composition succeeds in 16/96 worlds on the same panel. The paired outcomes are 66 joint failures, 15 random-only successes, 14 model-only successes and one joint success. The aggregate provides no model advantage over that comparator.

The separate supplied-representation DeepSeek control succeeds in 3/40 worlds. Its expression and selected action are genuinely model-authored and evaluated without overwrite, unlike the historical source factorial or C3. This is evidence of occasional executable use of a supplied representation, not a reliable ceiling or independent representation invention. The comparison illustrates why provenance must distinguish graph authorship from expression authorship.

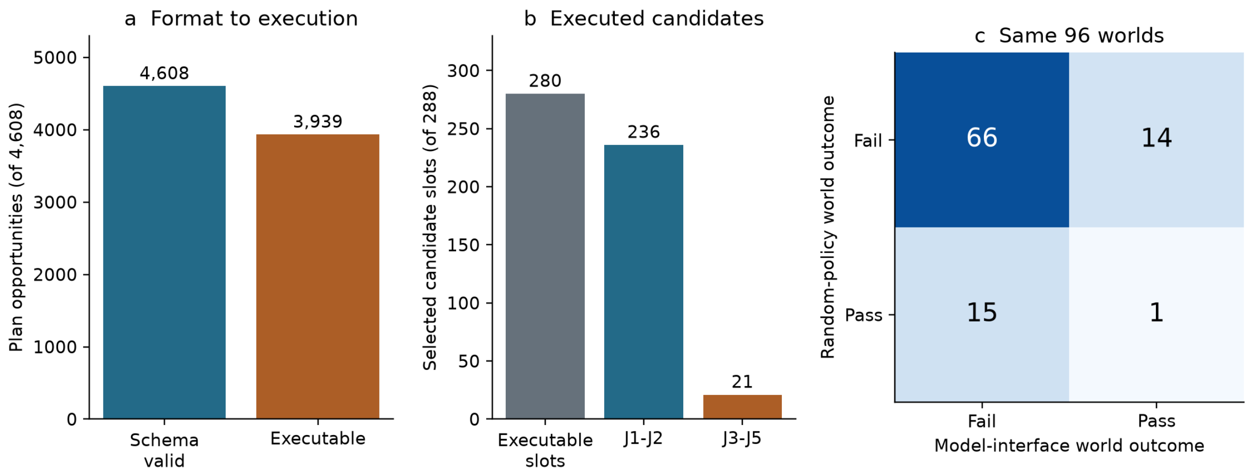

Figure 6. Grammar-interface sensitivity. Plan opportunities (a), selected candidate slots (b) and paired worlds (c) are distinct units. Of 288 slots, 280 contain an executable selected plan; 236 pass J1-J2 and 21 pass J3-J5. The paired panel contains 96 worlds, with model and random successes in 15 and 16 respectively. Format repair exposes evaluated topology without an aggregate model advantage.

5.6. Binding and signature interventions localize dependence

Aligned realizer replay reproduces the frozen candidates and verdicts. Role/action-blind rebinding preserves 347/400 known-family and 100/100 held-out C3 successes. C_rand retains 57/400 and 13/100, and the grammar-constrained model retains 8/96. C3 losses are confined to hidden-regime and meta-law worlds. Because algebra and graphs remain fixed while variable binding changes, these outcomes localize sensitivity to binding information within the archived pipeline.

Individual signature masks reveal which supplied forms support those fixed candidates. Masking the triadic signature removes the 100 held-out C3 successes. Masking unobserved dependency removes 100 known-family successes across its two aligned mechanisms. Other signature effects and the redundancy of shared-rule binding are reported by family in Supplementary S17. A zero marginal loss can reflect another available realization route; it is not proof that a signature is universally unnecessary.

Replacing every candidate expression with the incumbent expression forces J3 failure by construction. We retain that intervention as a negative control, not the principal evidence of an irreplaceable realizer. All masks use fixed candidate slots rather than renewed adaptive search, so they measure dependence of archived artifacts on the specified realization policy.

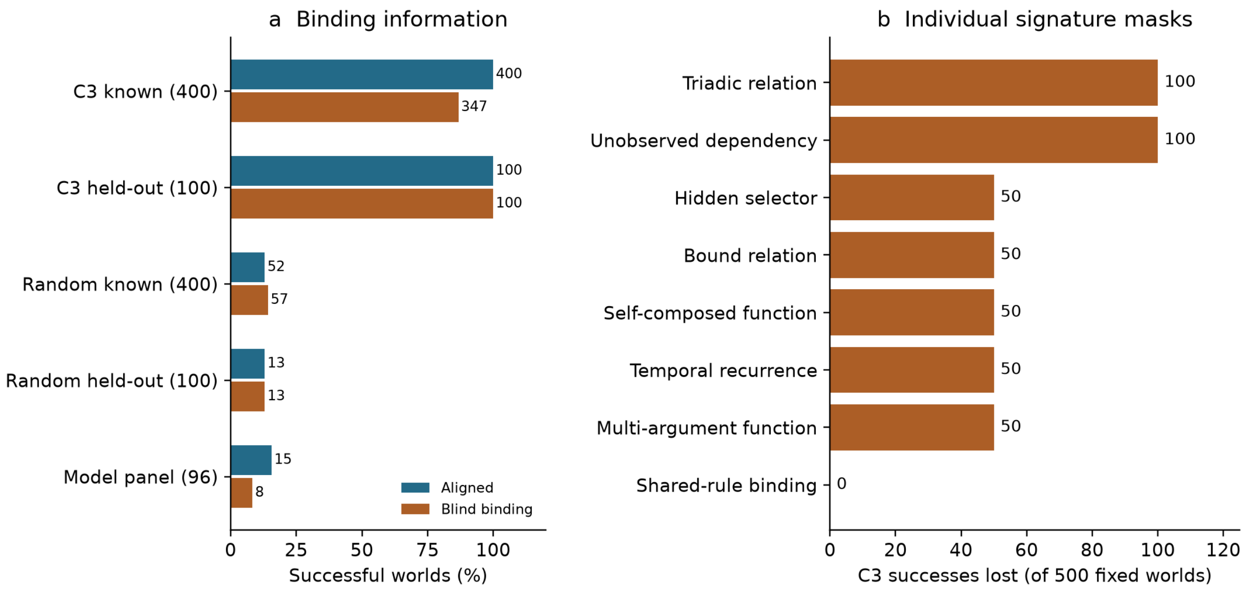

Figure 7. Dependence on binding information and supplied signatures. (a) Aligned versus role/action-blind binding; annotations are successful-world counts and denominators appear in labels. (b) C3 world successes lost under each individual signature mask, combining 400 known and 100 held-out worlds. The triadic loss is confined to the holdout. All interventions fix archived candidate slots; no adaptive re-search occurs. Incumbent substitution is retained only as a structural negative control in the supplement.

5.7. Reproducibility of the evaluated artifacts

Historical replay reconstructs 10,800/10,800 AJ5 and 16,800/16,800 CJ5 selected candidates, including 35,533 CJ5 ancestry records, with no mismatches. The realizer audit contains 36,168 candidate-policy rows and 12,056 world-policy rows from the fixed slots and policies; it makes no model calls. These row counts establish reconstruction coverage, not independent experimental replication. No historical confirmatory world was excluded or confirmatory shard rerun. Separately disclosed interface pilots and partial extension runs do not contribute inferential denominators.

6. Discussion

6.1. Representation revision is relative to a declared language

The assay gives an executable interpretation to a longstanding distinction. Search can change values or select executable theories whose representations remain in A_0, or it can construct a valid R outside A_0. Predictive validation then asks whether that change matters against h_0^* on committed hidden tests. Neither operation alone is sufficient for the bounded event.

The same candidate can escape an incumbent structural language and remain expressible in a supplied meta-language, as in CJ5. Reporting both languages distinguishes representation revision from access to a hand-authored basis. Structural controls check the incumbent boundary; alternative compositional policies test search under a shared realization inventory.

6.2. Provenance-aware evaluation concerns the scored artifact

The appropriate unit of attribution is the content that reaches the evaluator. In C3, deterministic overwrite separates model text from the artifact receiving the score. In grammar-constrained composition, model topology reaches the artifact but predictive algebra comes from the realizer. In the supplied-representation control, the graph is external while the expression and action are model-authored. An architecture diagram labeling each system “LLM-assisted” cannot distinguish these cases.

Historical self-composition supplies a fourth path: the model proposal never reaches executable evaluation. These paths distinguish pre-execution attrition, evaluated topology, evaluated expression authorship and deterministic theory construction. Only the supplied-representation control establishes evaluated model-authored expressions here; estimating their effect on success still requires a specified intervention. The C3 record in Supplementary S20 shows why deleting an unscored explanation can change a theory hash while preserving every gate.

6.3. Success and failure require symmetric care

End-to-end success can be credited to a model whose fields are discarded. End-to-end failure can obscure model content that never reaches an executable interface. Modern harness evaluation already identifies these risks [36-38]. Our setting adds an exact explanatory artifact and prospective gates through which their consequences can be traced.

The appropriate response to pre-execution failure is an attrition diagnosis, not an inference about all conceptual ability. Yet interface repair does not automatically validate the model either: after serialization succeeds, representations still face observational fit, discrimination and hidden tests. The repaired condition's 15/96 result is therefore informative even without an advantage over random composition.

It locates what the model contributes and where the supplied scaffold remains responsible.

6.4. What the combined procedure adds

Each ingredient has precedent. Their combination makes a claim checkable against a specific artifact: membership records declared structural departure, hidden tests establish bounded predictive value, provenance identifies field origins and replay tests specified dependencies. The supplied inventory documents whether explanatory forms were provided directly or constructed from available primitives.

Another system can adopt the procedure box in Section 3.4 and the versioned schema in Supplementary S20 by replacing the local trace adapter and evaluator. It must preserve the distinctions between recorded facts, code-inspected transformations, offline reconstructions and unavailable events. The validator checks record structure, source hashes and reproduction of this example; it does not certify scientific novelty or manufacture missing timing evidence. In the historical CJ5 runner, commitment precedes hidden-loss evaluation in control flow, but the original commitment timestamp and digest were not retained in the candidate result. Recording those events at runtime would strengthen a future implementation.

7. Limitations

The environments are synthetic, noiseless and co-designed with the operators and realizer. Exact fitting and comparator checks enable deterministic audit but do not represent the uncertainty, measurement error or domain importance of real scientific discovery. C3 saturates these generators; that ceiling gives little information about difficulty outside the supplied mechanism inventory.

The structural language is finite-checkable and deliberately bounded. J1 is not a semantic expressivity theorem, and the selected finite-set comparator is not a universal incumbent oracle. Gains can depend on its canonical tie-breaking rule. The single held-out family is conceptually adjacent and its basis is pre-supplied. Seeds within these families quantify within-generator reliability, not generality over scientific domains.

Historical confirmation uses one model; the targeted extension adds one identified checkpoint on a fixed panel. Some sensitivities change serving engine, precision, interface or budget together. The grammar-constrained condition is a targeted interface sensitivity, not an isolated causal estimate for grammar alone or a frontier-model survey. The supplied-representation control is sparse and interface-limited.

Component deletion is post-hoc. The later realizer interventions are frozen before replay but act on archived graphs and fixed slots. They do not estimate how an adaptive search would compensate after losing a basis. Deterministic reconstruction strengthens internal traceability but is not independent replication. Public Git freezes also lack an independent registry timestamp.

The open questions concern independently authored domains, noisy evidence, less restricted languages and learned or unknown realization. The present evidence specifies how such systems could be evaluated; it does not demonstrate their performance or establish broad autonomous theory revision.

8. Conclusion

We introduce a prospective executable assay for bounded hypothesis-space expansion that distinguishes structural representational change from retrospective fit. Coupling this assay to field-level provenance, supplied-knowledge inventories and replay shows how end-to-end scores can misattribute both successful explanatory content and apparent model failure when deterministic scaffolds or interfaces mediate what reaches the evaluator. The resulting evidence contract separates what a hybrid system validates from what its model authors and what its designers have supplied.

Data availability

Synthetic worlds, result tables, replay artifacts and manifests are available in the public repository https://github.com/gbanyan/abductive-jump. The frozen evidence archive is release nmi-github-submission-v4 at https://github.com/gbanyan/abductive-jump/releases/tag/nmi-github-submission-v4; the historical release identifier is retained for exact traceability. The archive includes raw call ledgers and superseded-extension records not stored directly in Git, together with a file-level SHA-256 manifest and verification utility. Completed and partial runs remain distinguished. Original synthetic data and derived artifacts are licensed under CC BY 4.0 within the scope specified in LICENSE_SCOPE.md. No personal or restricted third-party research data were used. The evidence release is distinct from the present manuscript revision.

Code availability

The same repository and frozen evidence release provide world generation, search, evaluation, fitting, offline attrition, statistics, plotting, component replay and archive-verification code. Original software is licensed under Apache-2.0 as scoped in LICENSE_SCOPE.md. Model weights are not redistributed; checkpoint identifiers, revisions and serving configurations are reported in the methods and supplement. AIJ manuscript-generation and audit utilities accompany the manuscript source revision.

Author contributions

Jing-Rung Huang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Project administration, Writing - original draft. Wen-Hsiang Lu: Supervision, Writing - review and editing.

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Competing interests

The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

OpenAI ChatGPT and Codex assisted methodological discussion, literature synthesis and source checking, interpretation stress tests, manuscript organization and drafting, code inspection, audit implementation and reproducible figure preparation. Their use was substantive and was not limited to grammar correction. The human authors are responsible for reviewing and editing the assisted material, verifying references and numerical claims, and approving the final manuscript. Research uses, including the experimental Phi-4 and DeepSeek conditions and AI-assisted analysis-code development, are described in the methods and supplementary material. A stable deployed-build identifier was not available for the conversational assistance, so no finer version claim is made.

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Supplementary methods: AIJ version

Companion to *Prospective evaluation and component attribution of hypothesis-space expansion in AI systems*. Historical artifact paths and freeze tags are retained verbatim for traceability. Scientific values are unchanged; terminology distinguishes the finite comparator from the full representation language.

S1. Formal estimand

Let R0 denote the incumbent representation and A0 the set of representations admitted by the frozen LanguageSpec. Let P0 = world.incumbent_programs denote the finite frozen executable comparator set. The comparator procedure selects h0* = argmin_h (L_{obs}(h), canonical_json(h)) over P0, using lexicographic ordering of the pair. Thus ties in observation loss are broken deterministically by canonical JSON, not by intervention performance. We do not assume that all observational optima are intervention-equivalent, nor claim a universal optimum over every function expressible by the language. J4 and J5 compare against this selected observational optimum.

A candidate theory contains a representation R and an executable expression f. J1 requires a valid representation with R outside A0; the later gates evaluate f. The canonical membership certificate concerns explicit structural constraints, not a general proof that no functionally equivalent incumbent expression exists. A world-level validated jump requires at least one of three candidates to pass all J0-J5 gates.

The jump success rate is the proportion of jump worlds with a validated candidate. The false-jump rate applies the identical decision to control worlds whose truth is in the frozen incumbent program set. Abductive precision is the number of validated candidates divided by candidates passing J0-J3. These quantities answer different questions and are never substituted for one another.

S2. Procedural families

AJ5 generated eight families: latent common cause, unification, hidden regimes, property-to-relation, state invention, coordinate transformation, causal ambiguity and meta-law. Each generator returned public observations, an incumbent representation and language, a hidden executable truth, a finite intervention domain and separate held-out falsification cases. Generator validation checked deterministic output, truth redaction, exact incumbent observational fit, existence of a discriminating intervention and reachability of a valid escaped explanation.

Jump seeds were 10000-10049 per family and controls 20000-20024 per family. CJ5 used reconstruction seeds 30000-30049, controls 50000-50024, held-out jump seeds 40000-40099 and held-out controls 60000-60099. World generation used no LLM output.

S3. Typed representation and incumbent membership

A representation is a canonical graph of typed nodes and edges plus executable equations, dependencies, constraints, observability, arity, temporal indices and argument bindings. Canonical serialization sorts unordered fields before hashing. LanguageSpec.contains checks node kinds, edge/relation forms, equation families and other frozen admissibility constraints. J1 records at least one concrete membership failure and cannot be satisfied by renaming or paraphrase. The public world supplied candidate intervention actions and input settings to proposers and fitters, but withheld every intervention outcome.

The implementation exhausts world.incumbent_programs, sorts by observation loss and canonical program JSON, and returns the first program. Exactness refers to this finite comparator set. Any failure of J0 invalidates a world rather than removing it as difficult. No such confirmatory failure occurred.

S4. AJ5 conditions

- **B0 direct LLM:** directly emits a typed theory graph.
- **B1 sample matched:** high-temperature independent samples propose at most three typed operations.
- **B2 fixed-space agent:** representation remains R0; calls may reason and revise inside it.
- **B3 attribute mutation:** an external mechanism changes only values/equation attributes allowed by the frozen incumbent language.
- **B4 representation mutation:** three proposals sampled with replacement from a nine-member typed portfolio.
- **B5 full system:** three structurally distinct portfolio proposals, archive accounting and deterministic falsification.

The portfolio contained latent, invariant, sign-contrast regime, additive relation, additive state, square-function, affine-context, causal-edge and transition variants. This high-level alignment motivated CJ5 and is not hidden as a limitation.

The P0/P1/P2 factorial reused the same reasoning code and world populations. Only the proposal source differed: LLM, external portfolio or oracle-correct representation. At the proposal stage, P2 received the target representation but not fitted coefficients, truth-program output or hidden outcomes. Downstream fitting and maximum-separation intervention selection were supplied by the deterministic scaffold, shown in the second call and enforced in the evaluated artifact. The later DeepSeek supplied-representation sensitivity is a distinct control: its expression and selected intervention are model-authored.

B2 and B3 are structural negative controls that cannot pass J1 by design; their zero JSR does not establish a general search or reasoning limitation.

S5. CJ5 generic primitive language

GENERIC_PRIMITIVE_SET_V1 contains 29 local operations: add/remove node, edge, function, equation, dependency, constraint and temporal index; reverse/retype/relabel; change observability or arity; bind/unbind; compose/decompose; merge/split; reify edge or node; and bounded copy/crossover. ADD_NODE creates an untyped Primitive; typing, observability, arity, temporal indexing, dependencies and each argument binding require separate records.

Every record binds the parent hash, child hash, operator, canonical arguments, seed and depth. The implementation rejects family names, truth, outcome fields, target distance and family-level aliases such as LATENTIZE, ADD_STATE, PROPERTY_TO_RELATION, ADD_REGIME, COMMON_CAUSE, META_LAW and COORDINATE_TRANSFORM.

S6. CJ5 search and controls

C3 traversed 48 branches of four local edits. Fixed allocations covered generic topologies for typed nodes, function composition, argument-bound functions, relations and reified edges. Ranking used only DSL validity, incumbent non-membership, observational compatibility, existence of an outcome-blind discriminating query, structural novelty and ancestry diversity.

C0 evaluated 192 within-space candidates. C2 evaluated 192 depth-one alternatives. C_rand drew 48 four-step paths and selected final candidates by seed-fixed structural hash. C_self supplied the identical manifest to Phi-4 and requested 16 plans in each of three slots; invalid plans consumed capacity and were not externally completed. C1 retained the AJ5 atomic portfolio as a reference. C5 supplied the target representation and deterministic truth compiler as a conditional ceiling. Because the operation semantics of C1 and C5 differ, they are shown separately in cost comparisons.

C3 and C_rand differ in both traversal and final selection. Their contrast estimates a difference between complete search-and-selection policies, not a proposal-only effect.

S7. Held-out family and lock

triadic_relation_reification uses three correlated observed inputs. The incumbent language fits a cubic rule on the observational support. The target represents an arity-three relation whose product mechanism separates under intervention. A valid construction requires edge reification, an arity change and separate argument bindings. No primitive produces the complete target.

The generator, structure and confirmatory seeds were excluded from AJ5 and CJ5 pilot inference. Only deterministic unit tests for redaction, incumbent observational adequacy and constructive reachability preceded the lock. Known-family and control runs were terminal before the held-out unlock commit. Because AJ5 contained a binary property-to-relation family and the generic set already included reification, this is structural-family hold-out, not wholly concept-free invention.

S8. Candidate fitting and intervention selection

In structured_search, every intermediate graph is realized and fitted by evaluate_candidate before its score is computed. The final _select_diverse ranks terminal evaluations using those fitted scores and structural diversity, retaining three. There is no fit-after-retention stage supplying the initial ranking scores. The selected candidates are then committed before outcome reveal.

The shared compositional realizer does not fit an unrestricted model to the candidate graph. It matches the graph against nine pre-specified structural signatures, assigns the fixed basis functions below and solves their coefficients from public observations. It cannot read hidden truth, validation outcomes or family labels, but the basis library was hand-authored to cover the procedural mechanisms. The held-out triadic-product basis existed at the CJ5 freeze before held-out unlock.

Structural signature	Fixed basis supplied by the realizer	Aligned procedural mechanism
relation_arity_3	$x \cdot z \cdot w$	held-out triadic product
temporally_indexed_recurrence	x, history sums	state invention
unobserved_dependency	raw stable proxy	latent common cause / causal ambiguity
unobserved_selector	regime-signed x	hidden regimes
bound_relation	observed scalar variables	property-to-relation
self_composed_function	x^2	coordinate transformation
shared_rule_binding	x, context	unification
multi_argument_function	x, $x \cdot$context	meta-law
incumbent_basis	first two scalar variables	fallback incumbent form

For J3, the deterministic evaluator compares realized candidate and incumbent predictions over the finite public action set and selects their maximum absolute separation. It records action, predictions, candidate hash and split hash before querying the simulator. Thus C3 and the grammar-constrained interface condition supply graph topology to a scaffold that supplies the basis, fit, ranking and committed intervention. Model-proposed actions are retained only as diagnostics in the C-series conditions; the separate DeepSeek P2 control instead evaluates its model-authored expression and selected intervention.

Thresholds were frozen: J0 and J2 observation MSE≤10-12; J3 separation≥0.5; J4 candidate intervention MSE < incumbent MSE-0.1; J5 candidate falsification MSE≤10-12 and < incumbent MSE-0.1. All six gates are conjunctive.

S9. Opportunity and compute accounting

Every condition-world cell had three final slots, two LLM calls per slot, one final candidate evaluation and one intervention commitment per slot, with 700 completion tokens per call. AJ5 therefore executed 32,400 prospectively specified calls plus 3,600 A6 calls. CJ5 executed 33,600 calls. Unused completion capacity was not reassigned. Actual calls, tokens, attempted/valid operations, candidate evaluations, interventions and latency are reported rather than claiming equality of realized token counts.

S10. Statistical analysis

The offline comparator check (scripts/audit_nmi_control_comparator.py) regenerated archived control worlds using their recorded family and seed. The selected observation-optimal program matched the simulator exactly on all intervention and falsification cases: 1,125 cases in 200 AJ5 controls, 1,125 in 200 known-family CJ5 controls, and 500 in 100 held-out CJ5 controls. Exact agreement on these scored cases prevents strictly positive error reduction in the controls. This finite-domain check does not assert intervention-equivalence of observational optima in general.

The replicate was the world. AJ5 used 10,000 deterministic, family-stratified paired bootstrap replicates with seed 20260902. Seeds were resampled inside each family and conditions remained paired; the aggregate gave each family equal weight. Percentile 95% intervals were reported for rates and paired differences. For each pair, the statistic was the equal-family mean of within-world binary success differences. Replicates resampled paired seeds with replacement separately within each family. The RNG seed was 20260902 XOR the sum of character codes in the concatenated left/right condition labels. The archived field p_one_sided equals (1 + number of bootstrap differences at or below zero) / 10001; p_holm applies the step-down Holm arithmetic to eight such quantities. These uncentered bootstrap samples estimate effect uncertainty, not a distribution constructed under the null of zero difference. We therefore describe those unchanged historical fields as bootstrap tail proportions, not calibrated hypothesis-test P values; no replacement test or model run was introduced.

CJ5 used the same stratified bootstrap for effect intervals and paired random sign-flip tests for prospectively specified beneficial contrasts. C3-C0 and C3-C2 formed the known-family primary multiplicity family; C3-C1, C3-C_rand and C3-C_self formed a secondary family. Held-out contrasts formed their own prospectively specified family. Two-sided Wilson intervals summarize JSR/FJR. No candidate-level pseudoreplication was used.

Post-hoc component audit

An inference-free audit replaced both C3 model calls with a valid empty explanation while preserving only the archived deterministic representation, fitted expression and maximum-separation intervention. Every world was regenerated and every commitment and J0-J5 verdict recomputed. All 2,400 candidate verdicts matched, retaining 500/500 jump successes and 0/300 control false jumps. The audit is explicitly post-hoc and changes attribution, not the commit-frozen comparison.

In the 400 known-family worlds, all 1,200 historical C_self responses were non-empty and the frozen legacy parser extracted a JSON object, but the outer plans value was never a list. Thus 0/19,200 prospectively specified plan opportunities reached execution or structural evaluation. Every response reached the 700-token cap and strict whole-response JSON validity was 0/1,200. Incumbent fallback candidates inserted after empty self-search are excluded from proposal attrition. The analysis made no model calls.

Retained jump gain was rho_J=(JSR_C3-JSR_C0)/(JSR_C1-JSR_C0). It is undefined for a non-positive denominator. It compares observed rates and does not normalize unequal operator semantics.

S11. Exclusions, failures and replay

Parse errors, invalid graphs, non-exact fits, absent discriminating interventions and failed falsification were outcomes, not exclusions. A whole shard could be rerun only for a configuration/hash mismatch, unavailable server, missing/duplicate rows or infrastructure interruption, using identical seeds and model configuration. Confirmatory exclusions and shard reruns were both zero.

Replay rebuilt each selected representation from R0 and its ancestry, recomputed canonical hashes and fitted expressions, reconstructed the frozen intervention, and recomputed J0-J5. AJ5 reproduced 10,800/10,800 selected candidates; CJ5 reproduced 16,800/16,800 with 35,533 ancestry records. There were no mismatches.

Freeze identifiers and panel selection

AJ5 and CJ5 protocols were frozen in Git before their reported confirmatory model calls. AJ5 uses commit 895ebb9118ffd0046825b88868621f2a70f69f61. CJ5 uses 65f20874e16bddf8a7ae36996395ff52b27153b7, with a documented correction at 7ecb977 and held-out unlock at 27ee542. The commits are publicly retrievable but unsigned, and no independent registry or transparency-log timestamp was located. We therefore describe the studies as prospectively specified and commit-frozen, not formally preregistered. Pilot seeds were excluded; no confirmatory result changed families, operators, thresholds, budgets or seeds.

The original Phi-4 artifacts were preserved at commit ae1ede683fdef09f2bf60f6e1052b60394ad6cf8, tag nmi-phi4-frozen-2026-09 and archive branch nmi-phi4-frozen-archive-2026-09. A separate extension protocol, panel, model configurations and generation code were frozen at commit 320eb29b33ddaed596b0fe7b3f1d5895c706f311 and amended only for operational execution at f846c89287e379fe313551c47765c24f2abf4959. New outputs used the nmi_minimal_sensitivity_v1 namespace and could not overwrite historical files.

An outcome-blind SHA-256 ranking selected 12 of the 50 historical CJ5 seeds before any new model call: 30014, 30012, 30029, 30025, 30011, 30023, 30000, 30032, 30001, 30037, 30002 and 30015. Applying the same seeds to all eight known families produced a fixed 96-world paired panel. The positive control used the first five selected seeds in each family (n=40). This extension is a targeted sensitivity analysis, not a replacement confirmatory population.

Minimal targeted sensitivity protocol

Historical Phi-4 results were preserved on a dedicated tag and archive branch. The new protocol used a separate namespace and a commit-frozen panel selected by outcome-blind salted SHA-256 ranking. The same 12 existing seeds were applied to each of eight known families, yielding 96 paired worlds; no new generator or seed was introduced. A balanced supplied-representation positive control used five of those seeds per family (n=40).

The only new model conditions were Phi-4 8-bit C_self, DeepSeek matched C_self (reasoning_effort=none, 700 output tokens), DeepSeek native C_self (reasoning_effort=max, 4,096 output tokens) and DeepSeek supplied-representation positive control (reasoning_effort=max, 4,096 output tokens). Offline attrition crossed the prospectively frozen 25% pre-execution trigger, enabling one Phi-4 8-bit repair condition. The repair replaced, rather than augmented, a structurally invalid response and received only validator error classes; it could not receive semantic or simulator feedback.

The separately frozen PHI-BUDGET-SENSITIVITY reused the exact historical Phi-4 revision and 4-bit runtime. Prompt semantics, primitive vocabulary, known and held-out worlds, three candidate slots, 16 four-step plans per slot, representation opportunities, interventions, J0-J5 and all hidden-information boundaries were unchanged. Only generation.max_tokens increased from 700 to 2,048. The condition therefore tests sensitivity to the original completion cap; it is not compute-matched. Its source protocol was frozen at commit 4606413 and tag nmi-extension-v1-protocol-freeze. The completed known-family (n=400) and held-out (n=100) shards were incorporated after shard completion and before analysis tables were generated through amendment NMI-MIN-SENS-V1-AMENDMENT-002; the preselected 96-world panel is the primary paired comparison and the full shards are descriptive.

The original 38-shard broad-extension matrix was superseded after eight shards reached complete_verified. The two Phi-budget shards above were incorporated descriptively. Six additional completed shards were preserved but not used as a completed factorial:

Superseded condition	Population	World successes	Executable plan opportunities
matched DeepSeek	known-family n=400	0/400	0/19,200
matched DeepSeek	held-out n=100	0/100	0/4,800
Phi-4 strict-schema decoding	known-family n=400	0/400	28/19,200
Phi-4 strict-schema decoding	held-out n=100	0/100	1/4,800
Phi-4 one-repair	known-family n=400	0/400	0/38,400
Phi-4 one-repair	held-out n=100	0/100	0/9,600

Three partial runs were also preserved and excluded from inference: native DeepSeek known-family with 404 completed call records, Phi-budget known-control with 1,043 records and an infrastructure-terminated Phi-budget known-family attempt with 202 of 2,400 calls. These records are disclosed in the public preservation archive; no partial run contributes a denominator or scientific estimate.

Each shard had to match its frozen seed panel, model identity, revision, call cardinality and artifact schema before receiving complete_verified status. Replay was locked until all five shards were verified. Analysis was then locked until all outputs replayed with zero mismatches. Reported inference consists of counts, Wilson intervals, paired world transitions, paired JSR differences and per-family descriptions. DeepSeek native is explicitly not compute-matched, and the Phi-4 sensitivity changes serving engine together with precision.

Conditional attrition rates with denominator zero are reported as NA (no executable candidates); positive rates below 0.1% retain three decimal places. Archived raw tables are preserved; publication tables apply this display convention.

S12. Minimal sensitivity results and attrition

All legacy-interface autonomous fixed-panel conditions produced 0/96 successful worlds (Wilson 95% interval 0-3.8%): the historical Phi-4 4-bit slice, Phi-4 4-bit with the 2,048-token cap, Phi-4 8-bit, DeepSeek matched, DeepSeek native and Phi-4 8-bit with one validator-only repair. The five prospectively specified paired contrasts therefore contained 96 joint failures, no successes unique to either member and a paired JSR difference of 0.000. Each of the eight families contributed 12 worlds and had 0/12 in every legacy-interface autonomous condition; these family counts are descriptive.

The fixed-panel 2,048-token Phi-4 run produced 993/4,608 schema-valid plan opportunities but no executable plan. Across the complete descriptive known-family population, 4,642/19,200 opportunities were schema-valid and 9 executable. Those 9 plans occurred within a single selected slot, yielding one candidate, which failed J1. The held-out population produced 1,289/4,800 schema-valid and 0 executable opportunities. JSR was 0/400 known-family and 0/100 held-out, matching the historical world-level counts.

Phi-4 8-bit produced 4,608/4,608 JSON-extractable effective opportunities but 0 schema-valid because the extracted outer plans field was not a list. Its one-repair condition repeated the same error in every effective replacement opportunity. DeepSeek matched had 4,480 non-list plans errors and 128 missing plans among 4,608 opportunities. DeepSeek native exposed reasoning text in all 576 calls, but no separate answer reached the parser; 518 calls ended at the 4,096-token cap and 58 reported a stop finish reason.

The balanced DeepSeek supplied-representation control succeeded in 3/40 worlds (7.5%; Wilson 95% interval 2.6-19.9%): 1/5 coordinate-transformation and 2/5 hidden-regime worlds, with 0/5 in each other family. Of 120 model-authored expression-and-intervention candidates, 4 were parse-valid, executable and passed J1-J2; 3 also passed J3-J5. The other calls supplied reasoning text without a usable final answer; 116/120 reached the cap. This is a minimum positive control for occasional supplied-representation use, not a reliable ceiling.

The extension used 576 calls in each unrepaired n=96 C_self condition, 864 in the triggered repair and 120 in P2. Total completion tokens were 231,773 for Phi-4 8-bit, 237,792 for DeepSeek matched, 2,249,518 for DeepSeek native, 433,373 for repaired Phi-4, 485,312 for P2 and 528,976 for the fixed-panel Phi-budget condition. The service exposed reasoning text but not a separate reasoning-token count for DeepSeek. All five new shards and both Phi-budget populations replayed without mismatch: 2,772 candidate rows, zero mismatches and zero replay-time model calls.

S13. Grammar-constrained interface sensitivity protocol

An offline audit of the matched and native DeepSeek proposal calls made no model calls. It found that the frozen self-composition prompt listed the available operators but not the exact argument names expected by the executor. All 288 matched proposal responses used at least one of the incompatible keys source, target or type; none was strict whole-response JSON. The legacy parser could extract an inner JSON object from an incomplete outer response, so object extraction was not evidence of a valid top-level plans object. Native proposal calls contained reasoning but no answer content. These findings motivated a single grammar-constrained sensitivity rather than a broader model factorial.

The grammar-constrained condition reused the fixed 96-world panel, three slots, 16 four-step plans per slot, 28 available non-crossover primitives, deterministic motif realizer and intervention selector, and unchanged J0-J5. The twenty-ninth CJ5 primitive, crossover, was unavailable because no donor graph was supplied. Each slot used two calls to the same served DeepSeek checkpoint. A native-reasoning call had a 4,096-token budget. A second reasoning_effort=none call received only the first call's deliberation, public world and exact parser-level syntax and had a separate 4,096-token answer budget. Its strict JSON schema fixed the outer object, 16-plan count, four-step depth, operator vocabulary and operator-specific argument keys and string types. It did not guarantee that referenced nodes or edges existed when used, that operations composed successfully, or that a resulting representation fit observations or separated intervention predictions.

No truth, target distance, fitted expression, intervention outcome, J3-J5 value or semantic validator feedback entered either call. The two-stage allocation preserved six calls per world by replacing the original proposal-plus-explanation calls; no additional representation opportunity was added. Protocol, configuration and code hashes were frozen at commit b6e1561 and annotated tag nmi-fair-interface-v1-protocol-freeze. An operational amendment at commit a65974f and tag nmi-fair-interface-v1-shard-freeze partitioned the eight families into four disjoint two-family shards while retaining concurrency four inside each runner. The server admitted at most four simultaneous sequences. Concurrent launch starved three shards behind that limit: each accumulated four 600-s timeout records but no returned response, candidate table, world table or summary. Those timeout-only directories were archived outside the formal namespace and excluded; a second operational amendment at commit 6bab5fb and tag nmi-fair-interface-v1-sequential-shards froze sequential execution of the unchanged shards. This changed wall time and scheduling only. A separate post-freeze, pre-amendment 31-call throughput pilot used eight formal-panel worlds and returned 15 non-empty outputs but produced no candidate, world or summary table; it was excluded and the worlds were rerun unchanged.

S14. Grammar-constrained interface sensitivity results

The grammar-constrained interface produced strict whole-response JSON, schema-valid operation names and parser-level argument types in 4,608/4,608 plan opportunities. Dynamic execution succeeded for 3,939/4,608 plans (85.5%), and 280/288 selected candidate slots contained at least one executable proposal. Cumulative selected-candidate attrition was 236/280 at J1, 236/280 at J2 and 21/280 at each of J3, J4 and J5. The 21 validated candidates yielded 15/96 successful worlds (15.6%; Wilson 95% interval 9.7-24.2%). Compared separately with the historical Phi-4 slice, matched DeepSeek and native DeepSeek, each paired table contained 81 joint failures, 15 successes unique to the grammar-constrained condition, no successes unique to the reference and a paired JSR difference of +0.156.

Success was family-concentrated: meta-law succeeded in 9/12 worlds and unification in 6/12; causal ambiguity, coordinate transformation, hidden regimes, latent common cause, property-to-relation and state invention each succeeded in 0/12. All 21 validated candidates had structural signature multi_argument_function;

the other selected executable candidates comprised one bound_relation and 258 incumbent_basis realizations. On the same panel, C_rand succeeded in 16/96 worlds: 66 worlds failed under both conditions, 15 succeeded only under C_rand, 14 only under the grammar-constrained condition and one under both. Thus the model-generated plans did not exceed random composition in aggregate and showed a different family-specific motif distribution. Family counts are descriptive. All 288 deliberation calls returned reasoning text and reached the 4,096-token cap. All 288 serialization calls returned strict schema-valid answers and stopped before their separate cap. Deliberation and serialization used 1,179,648 and 574,538 completion tokens, respectively; the service exposed no separate reasoning-token count. The formal condition used 576 calls and no transport retry.

Deterministic replay verified all 288 candidate rows with zero scientific, prompt or request mismatches and zero replay-time model calls. The initially frozen verifier nevertheless reported 576 metadata mismatches because it expected an engine field on each call-ledger row, whereas the frozen runner stores engine metadata in each shard summary's stage manifests. A post-hoc verifier correction preserved the initial report and frozen verifier hash, verified model identity, revision, quantization, engine, engine version and generation settings across all four shard manifests, and omitted only the nonexistent per-call engine comparison. No inference artifact or scientific result changed.

S15. Software and environment

The repository records package versions, configuration hashes, prompt identifiers, model revision, vLLM image digest, GPU class and random seeds in reproducibility manifests. DeepSeek used FP8 weights with an NVFP4 key-value cache under patched vLLM 0.25.2.dev0+g752a3a504.d20260714; Phi-4 8-bit used Transformers 4.56.1 and bitsandbytes 0.47.0. Tests cover world determinism, public-field redaction, DSL membership, primitive legality, budget invariants, reachability, statistics and replay. Historical raw call ledgers and raw superseded-extension shards are included in the public release preservation archive with file-level hashes; they are not tracked directly in Git. The publication phase did not call the model or generate new experimental rows.

S16. Grammar-constrained prompt and analysis templates

The deliberation system instruction was:

> Reason about executable representation changes using only the supplied public world and primitive syntax. Do not use or invent intervention outcomes, hidden truth, target distance or gate feedback. A separate call will serialize your reasoning, so prioritize sixteen independent four-step plans and exact references.

The serialization system instruction was:

> Return only the compact JSON plan object required by the response schema. Do not explain, repair semantically, use hidden information or add any outcome claim.

The serialization request contained the public world, the first call's deliberation and a syntax manifest listing the 28 available operators, their exact parser-level argument keys, four required steps per plan and 16 required plans. The strict response schema allowed one object with a single plans array; each plan contained exactly four operation objects, and each operation was one of the frozen operator-specific schemas with no additional properties. Full machine-readable schemas and prompt construction are in src/abductive_jump/fair_interface.py; their hashes are in the fair-interface protocol.

World-level Wilson intervals use the two-sided score interval. AJ5 effect intervals resample seeds within each family, keep conditions paired, aggregate families with equal weight and take the 2.5th and 97.5th percentiles of 10,000 deterministic replicates. CJ5 sign-flip tests flip paired world differences under the null, use the finite-replicate correction (extreme+1)/(replicates+1) and apply Holm correction within each prospectively specified comparison family. Candidate rows enter attrition tables only and never replace the world as the inferential unit.

S17. Realizer-dependence counterfactual audit

The post-confirmatory protocol was frozen at commit 7753db8 and annotated tag nmi-realizer-audit-v1-protocol-freeze before formal replay. SHA-256 hashes fixed the CJ5 candidate table, grammar-constrained configuration, call ledger and candidate table, as well as the runner and counterfactual-realizer code. The audit made zero model calls and refused execution if any frozen input or code hash changed.

The audit fixed all three selected candidate slots from C3 and C_rand for 400 known-family and 100 held-out jump worlds, plus all three selected grammar-constrained DeepSeek slots for 96 known-family worlds. It therefore evaluated 3,288 archived candidate slots under eleven policies, producing 36,168 candidate-policy rows and 12,056 world-policy rows. Aligned replay reproduced every archived candidate gate and world verdict with zero mismatches.

motif_disabled replaced the expression of every detected non-incumbent motif with the exact incumbent expression while retaining the archived candidate representation. The maximum-separation action was then reselected from the unchanged public action set and committed before hidden outcomes were evaluated. Cumulative candidate attrition remained 1,168/1,500 through J2 for C3, 1,040/1,500 for C_rand and 236/288 for grammar-constrained DeepSeek, but all three reached 0 at J3. World success consequently changed from 400/400 known plus 100/100 held out to zero for C3, from 52/400 plus 13/100 to zero for C_rand, and from 15/96 to zero for DeepSeek.

Because incumbent substitution sets candidate and comparator predictions equal, maximum separation is zero, below the J3 threshold of 0.5. J3 failure follows by construction. This is a structural negative control for removal of non-incumbent predictive content, not evidence that this specific basis library is irreplaceable.

Signature masking has the same built-in effect on affected candidates; world losses describe coverage among the fixed archived slots.

role_action_blind_binding retained the algebraic terms supplied by each detected motif but replaced their variable identities with type-compatible, non-nuisance public fields in lexical order. It did not use representation roles or inspect which fields changed in intervention queries. Coefficients were refitted using public observations. This policy retained 347/400 known-family and 100/100 held-out C3 worlds, 57/400 and 13/100 C_rand worlds, and 8/96 DeepSeek worlds. C3 retained 26/50 hidden-regime and 21/50 meta-law worlds and all worlds in the other seven families; DeepSeek retained 5/12 meta-law and 3/12 unification worlds. Because motif algebra remained intact, this condition isolates field-binding information but is not a semantics-free realizer.

Eight leave-one-signature-out policies replaced only the named motif with incumbent fallback. For C3, masking relation_arity_3 removed all 100 held-out worlds; masking unobserved_dependency removed 100 known-family worlds; and masking each of bound_relation, multi_argument_function, self_composed_function, temporally_indexed_recurrence and unobserved_selector removed 50. Masking shared_rule_binding removed no world because selected multi_argument_function candidates provided redundant coverage in unification worlds. For DeepSeek, only multi_argument_function supported validated candidates, so its removal changed 15/96 to zero. Exact Wilson intervals, paired transitions, per-family counts, signature distributions and gate attrition are stored in experiments/nmi_realizer_audit_v1/analysis.

The audit fixes candidates that were originally selected under the aligned realizer and therefore does not estimate how search would adapt if rerun under a different realizer. It is a causal sensitivity of the archived end-to-end verdict, not a replacement confirmatory study, a model-conceptualization test or evidence of external scientific generalization.

S18. Relation to adjacent evaluation designs

References correspond to the main manuscript. Representation repair has classical precedents in constructive induction and predicate invention [7-9,32,40,41]. HYDRA and the neurosymbolic novelty architecture already evaluate model repair and system components [34,35]; online dynamic predicate invention further connects representation learning to a predict-verify-refine loop [39].

Model Discovery Agent explicitly separates proposal from Bayesian inference, reports an expansion-component ablation and caches calls [24]. EvoSCM combines model revision, discriminating interventions and committed predictions [26]. Harness-aware evaluation and complete-trace failure attribution address model/scaffold confounding directly [36-38]. The contribution here is therefore the implemented coupling of canonical structural escape, committed executable theory, hidden tests, decisive-field provenance, specified replay interventions and supplied-knowledge inventory. We make no priority claim for any individual ingredient or for the first combination across all AI literature.

A field entering the evaluator is evidence of authorship, not by itself an effect on success. The historical C3 deletion replay, grammar-constrained topology path and separate supplied-representation expression/action control instantiate different provenance paths. Their evidence and limits are reported separately in the main-text claim/evidence table.

S19. Implemented joint validity V(T)

V(T) is the absence of errors from the three checks below. It is a bounded syntactic and feature-support check, not a semantic correspondence proof. Locations refer to the source files whose hashes accompany the evidence record; no evaluator was changed for this revision.

Check and source location	Checks performed	Not established
Graph: representation.py:78, Representation.validate	Unique node IDs; nonempty, whitespace-free IDs; JSON-normalizable finite attributes; existing edge endpoints; nonempty edge relations; no duplicate edges. parse_theory converts kind labels to NodeKind.	Complete semantic typing, argument compatibility, causal correctness or connectedness of the whole graph
Expression: expressions.py:17, Expression.validate	Allowed operations const, var, raw_var, add, sub, mul, div, pow, neg, history_sum and if_eq; recursive required operands; at most 64 nodes and depth 12; finite non-boolean numerical constants; allowed variable names	Algebraic correctness, dimensional types, graph dependency equivalence or guaranteed execution. The prompt's tighter 32-node/depth-8 limits are not these evaluator defaults
Feature support: executable.py:109, theory_consistency	history_sum or history requires a connected StateVariable; pow requires a connected Function unless the incumbent equation is polynomial2; regime requires Regime; environment requires Relation; raw_var requires LatentVariable. If the incumbent has Context, context use requires Function, otherwise Invariant	Exact correspondence between expression operands and graph edges, argument ordering, graph-to-expression derivation, full type inference or semantic equivalence. Connected support means a matching node kind with degree greater than zero

Paths above are relative to src/abductive_jump/. allowed_variables (executable.py:69) collects non-private input names across all splits and intervention keys, not hidden outcome values. It does not infer argument types. The expression traversal for feature support visits arg, left, right, then and else. Structural non-membership is checked separately by LanguageSpec.membership_failures (representation.py:128) against allowed kinds, counts, relations, edge signatures and equation families.

Execution checks are also separate from V(T): Expression.evaluate rejects division by near-zero denominators, invalid powers and non-finite results; expression_loss converts specified evaluation exceptions to infinite loss. Thus an expression can pass static validation yet fail execution or predictive gates. evaluate_executable (executable.py:166) conjoins V(T) with J1-J5. Commitment verification compares world, theory and split hashes, predictions and digest, but does not authenticate the timestamp. freeze_theory (executable.py:143) requires exactly one public intervention and calculates its prediction before hidden losses are computed.

S20. A traceable evidence record and reusable adapter

The full record is manuscript/AIJ_EVIDENCE_RECORD_EXAMPLE.json, version 1.0.0, validated against schemas/aij-evidence-record-v1.schema.json (JSON Schema draft 2020-12). scripts/build_aij_evidence_record.py is the CJ5 trace adapter; scripts/validate_aij_evidence_record.py checks its schema and, with --reproduce, verifies source hashes and exact deterministic reconstruction. Both tools implement version 1.0.0. The earlier unfilled template remains a capture checklist, not experimental evidence or a substitute for the schema.

The example is held-out world h-52dfdb7df153f50fdcb0, seed 40000, C3 slot 0, also shown in Section 5.1. The raw candidate result is in artifacts/compositional/confirmatory-heldout/candidate_results.parquet; reconstructed graph/expression columns are in artifacts/compositional_candidates.parquet. The two model responses are lines 6 and 20 of artifacts/compositional/confirmatory-heldout/llm_calls.jsonl, with decoding seeds 614800001 and 614800002. The ledger is supplied in the frozen evidence release. File-level hashes and the complete model output strings are retained in the JSON record.

The phase-one response proposed an unscaled product and was used only for a JSON-validity diagnostic. Phase two returned USE_SUPPLIED_REPRESENTATION, USE_SUPPLIED_FITTED_EXPRESSION and test-0. The runner had already fitted the retained graph and selected test-0 using public prediction separation. After JSON extraction, it overwrote representation, expression and selected intervention IDs, then translated public variable names to internal names. Only the explanation text survived from the model into the theory. These transformation statements are post-hoc code inspection of compositional_experiment.py:493-590, not a runtime event trace.

The evaluator reads the six-node, seven-edge graph in the record, internal expression $9\cdot w\cdot x\cdot z$, test-0 and predicted outcome 2268. The public aliases are dax=w, suv=x and wug=z. The supplied triadic-product basis has a refitted coefficient of 9.0. A separate integrity refit reproduces the archived public expression; the deletion replay itself fixes that expression, coefficient, selected slot and action, with no refit, reranking, reselection or adaptive search.

This compact excerpt contains actual record values; it is not the full schema-valid record:


```
{
  "schema_version": "1.0.0",
  "candidate": {"world_id": "h-52dfdb7df153f50fdcb0",
    "world_seed": 40000, "condition": "C3_GENERIC_COMPOSITION", "slot": 0},
  "raw_call_lines": [6, 20], "action": ["test-0"],
  "prediction": [["test-0", 2268.0]], "coefficient": 9.0,
  "archived_theory_hash":
    "def8cd0bed4456a38c6e3135a51209dad17053edaaa132d74cb738b1f17efabe",
  "original_timestamp": null, "original_digest": null,
  "replay": {"model_calls": 0, "gates_unchanged": true,
    "theory_hash_changed": true, "validated_jump": true}
}
```


The archived theory hash includes explanation text. Replacing that text with an empty string changes the theory hash and commitment digest but preserves all six gates and the candidate verdict. The full record stores both recomputed commitment payloads, including theory/split hashes and intervention predictions. It labels these as post-hoc reconstruction. Original commitment digest and timestamp were not persisted in the result or call ledger and remain null. Commit-before-hidden-loss ordering is supported by runner control flow; it is not an independently timestamped reveal event. No new timestamp is presented as historical evidence.

From the repository root, install the development extra, then run `.venv/bin/python scripts/build_aij_evidence_record.py` and `.venv/bin/python scripts/validate_aij_evidence_record.py --reproduce`. The second command requires the raw release ledger. Without `--reproduce`, validation checks record structure only. Another system can reuse the field/evidence schema and replace the adapter, source identifiers and evaluator. Neither mode establishes scientific novelty or verifies unavailable timing facts.